LESSON 08 · MISSION GUIDE

Whole Columns at Once

NumPy: do the maths on 200 salaries in one line, no loop, no dragging.

From **Mo, IT**
To **Jen**
Subject **RE: RE: RE: the loop**

Jen. I watched MACRO-9000 do the pay review on Friday. It loops over 200 rows, one at a time, recalculating the whole workbook between rows. Forty seconds. Sometimes it skips row 1, nobody knows why. Priya makes tea while it runs. She has a dedicated mug.

NumPy treats a whole column as one thing: `salaries / 12` divides all 200 at once. It's dragging a formula down, except it's instant and never skips row 1. Install it with pip, then go and retire Priya's mug. — Mo

What you'll be able to do after today

- [ ] Turn a list of numbers into a NumPy **array** and ask it for min, max, mean, median, sum
- [ ] Do **column maths** with no loop: `salaries / 52 / hours` gives 200 hourly rates at once
- [ ] Filter with **boolean masks**: `salaries[salaries > 40000]` is AutoFilter in one line
- [ ] Combine filters with `&`, `|` and `~`
- [ ] Compute **percentiles and quartiles**, the ones the pay gap report asks for
- [ ] Work with a 2-D array (a whole range) and sum it along either axis

Why this matters at work

- **This is the real pay gap calculation.** Exercise 4 computes Brightwell's mean and median gender pay gap from all 200 employees. Exercise 5 does the quartiles. Those are two of the six numbers the government wants.
- **pandas is built on NumPy.** Next lesson's DataFrame columns *are* arrays. Everything you learn today (masks, axis, vectorised maths) carries over directly.

Four big ideas

1. An array is a column	`np.array([1, 2, 3])`. One type for the whole column (`dtype`), one shape. Like a column of numbers in Excel, not a list of mixed things.
2. Maths is element-wise	`salaries * 1.03` raises every salary. `salaries / hours` divides row by row. No loop. That's "vectorised".
3. A comparison gives a mask	`salaries > 40000` is an array of True/False. Put it in square brackets to filter: `salaries[mask]`. `mask.sum()` counts the Trues.
4. Axis	In a 2-D array, `axis=0` works down the columns, `axis=1` along the rows. SUM at the bottom of a column vs SUM at the end of a row.

Three things that will bite you

- `and` / `or` don't work on arrays. Use `&` / `|`, and put brackets round each comparison.
- `np.percentile(x, 75)`, not `0.75`. Excel's `PERCENTILE` wants 0.75; NumPy wants 75.
- `salaries.mean` without brackets is the method, not the number. If you see `<built-in method mean>`, add `()`.

How today works

15 MIN
Lecture
Lars does the pay review in one line, then explains it

60 MIN
Exercise sheet
Seven tickets on the real 200-row payroll file

15 MIN
Dave's Leftovers #8
Dave's quartile report. Gerald has the error message

The inbox

#	FROM	TICKET	YOU'LL USE
1	Priya	Your first column	`np.array()` `.shape` `.min()` `.mean()` `np.median()`
2	Priya	Dragging a formula down	array maths, `.round(2)`
3	Siobhan	AutoFilter, but faster	masks, `&` `|` `~`, `.sum()`
4	Gerald	The gap, properly this time	masks by gender, mean vs median
5	Gerald	Pay quartiles	`np.percentile()`
6	Mo	A whole range at once	2-D arrays, `axis=`, `[row, col]`
7 ⭐	Gerald	What if?	`np.where()`, `np.maximum()`, a function

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if the board can wait), each printing the expected output
- [ ] Dave's quartile report runs and prints the expected three lines, with one commit per bug
- [ ] Everything is pushed to `lesson-08` and you've opened a pull request



Next lesson: The Spreadsheet That Isn't. pandas gives you a DataFrame: a whole table with named columns, filters, sorting and reading straight from Excel. It's the spreadsheet, without the spreadsheet.